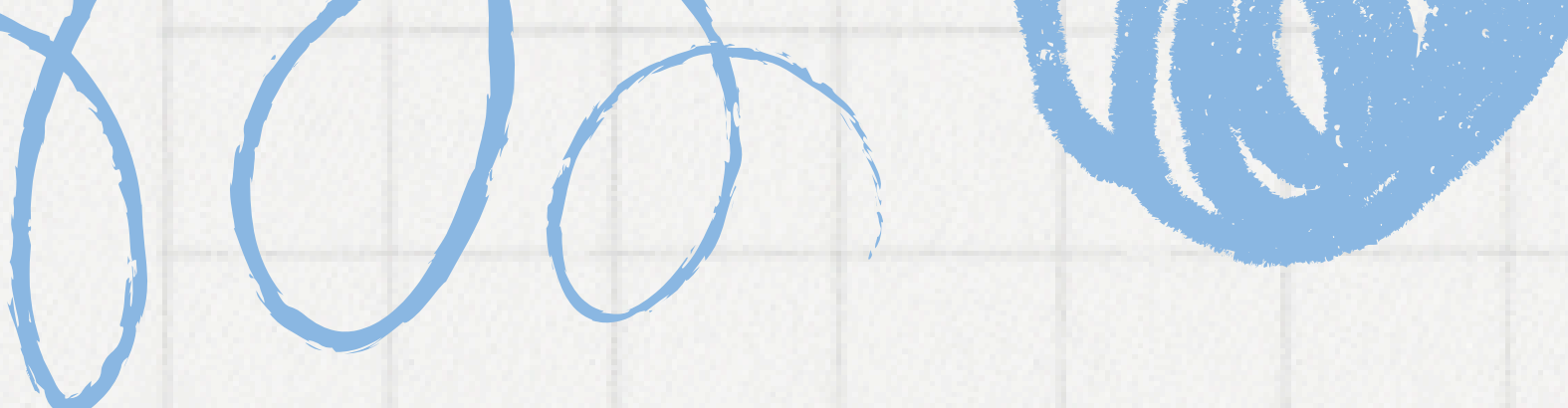




Mention-level Keyphrase Identification, Classification, and Semantic Relation Extraction



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Introduction

Our project aims to do three things: find important words or phrases in text, sort them into groups, and understand how they relate to each other, specifically for the domain of scientific publications.

- mention-level keyphrase identification
- keyphrase classification
- semantic relation extraction between keyphrases of the same type



SELECTION OF APPROPRIATE BASELINES

- In SemEval 2017 Task 10, The top three teams used an RNN architecture. Some other methods used were SVM and CRF.
- In another paper, Yi Luan et al. made a system called SCIE. It finds scientific expressions in text and connects them. They used neural networks to do this.
- We also tried using a transformer model for finding important words in text. This model is good at understanding context. The baseline method used in the paper was to randomly guess labels for each word.
- The paper showed the best possible score we could get. The paper also mentioned an upperbound on F1 score due to sentence splitting and tokenization methods.

Dataset Characteristics

Annotated Textual Data with Semantic Relations

- The dataset comprises textual data annotated with keyphrases and their classifications, alongside semantic relations.
- Semantic relations like hyponymy and synonymy are denoted within the annotations.

Multifaceted Domain Coverage

- The data is segregated into three folders: train, test, and dev, each containing .txt and .ann files for text and annotations, respectively.
- Additional .xml files from ScienceDirect provide background information, enriching the dataset but not essential for analysis.

Data Statistics and Categorization:

- Keyphrases are categorized into three classes: PROCESS, TASK, and MATERIAL, facilitating structured analysis.
- Moreover, the dataset includes annotations for HYPONYM-OF and SYNONYM-OF relations, offering insights into hierarchical and semantic associations among keyphrases.

Approach 1

- It involves using BiLSTMs (Bidirectional Long Short-Term Memory networks), a type of recurrent neural network, for both the subtasks.
- Approach 1 involves the sequential process of analyzing the text, identifying keyphrases using BiLSTM and then classifying these keyphrases into predefined categories using a BiLSTM classifier.

Approach 2

- In this approach, a pre-trained transformer model (in our case DistilBERT) is employed. The model is fine-tuned on both tasks simultaneously. This means the model learns to identify keyphrases and classify them into categories together.
- Approach 2 involves preprocessing the text, fine-tuning a pre-trained transformer model on the combined tasks of keyphrase identification and classification, and training the model using a multi-task learning objective. This approach uses the ability of transformer-based models to understand the contextual relationships



Steps for both the Approaches

Approach 1

01

Text Analysis and
Tokenization

02

Keyphrase
Identification using
BiLSTM

03

Keyphrase
Classification using
BiLSTM Classifier

Approach 2

01

Preprocessing and
Tokenization

02

Fine-tuning
Transformer Model
(DistilBERT)

03

Multi-task Learning

Results and Analysis

We evaluated the performance of our implemented approaches for F1-score, precision, and recall, for each task.

Table 1: Evaluation Results

Dataset	Approach 1			Approach 2		
	Precision	Recall	F1 Score	Precision	Recall	F1 Score
Validation	0.66	0.67	0.66	0.3386	0.4052	0.3690
Test	0.59	0.57	0.56	0.2929	0.3576	0.3220

Qualitative Analysis



Performance evaluation metrics show similarity in performance between classes I and B, as well as between Process and Material classes.

- In Keyphrase Identification, the O class tends to perform better than I or B classes due to its higher frequency in the dataset, causing the model to lean towards classifying tokens as O.
- Process and Material labels have a higher frequency compared to the Task label in Keyphrase Identification tasks.

In Relation Extraction, the "No Relation" class is significantly more frequent than other classes due to the absence of relations between most pairs of keyphrases.

Challenges Faced

Dependence on High-Quality Annotated Data

Obtaining high-quality annotated data can be expensive and time-consuming, limiting the scalability of the models.

Struggle with Domain-Specific Terminology

Models may face challenges in handling domain-specific terminology or rare keyphrases that are not well-represented in the training data.

The background is a light gray grid. It is decorated with various hand-drawn blue doodles. At the top, there are several overlapping circles and loops. On the right side, there are some star-like or burst-like shapes. At the bottom, there are more circles, a wavy line, and several small checkmarks or 'v' shapes.

Thank you very much!

Any questions?